EduAssist AI: API Cost, Latency and Prompt Version Management

AIMLCZG523 MLOps - Assignment 2 | Student ID: 2025AE05025 | 30 October 2026

1. Setup

EduAssist answers student questions in JSON with *answer*, *confidence* and *key_points*. I used **gemini-3.5-flash-lite** through the Gemini API v1beta OpenAI-compatible endpoint with the openai Python SDK 3.41.0, temperature 0. The API key is read from the GEMINI_API_KEY environment variable. Costs are calculated with the Standard paid-tier price of USD 0.30 per 1M input tokens and USD 2.50 per 1M output tokens (ai.google.dev pricing page, checked on 30-Oct-2026, USD). The calls themselves ran on the free tier. MLflow 3.3.2 with a SQLite backend was used for tracking.

2. Task 1 - Latency

Five academic questions were sent with the JSON prompt (v3) and timed with time.perf_counter(). All five succeeded.

Question	Latency (s)	Output tokens	Status
Q1 Overfitting	1.213	168	success
Q2 Supervised vs unsupervised	1.192	125	success
Q3 Binary search complexity	1.325	115	success
Q4 Primary key	1.007	116	success
Q5 Gradient descent	1.333	171	success

Mean 1.214 s, median 1.213 s, min 1.007 s, max 1.333 s, std 0.132 s. Q5 (gradient descent) was the slowest. It also had the longest answer (171 output tokens), because the question asks for a simple explanation and the model generates the answer token by token. Output length is not the only factor: Q3 had only 115 output tokens but took 1.325 s. Network delay and load on the shared free-tier endpoint add a few hundred milliseconds of variation.

3. Task 2 - Tokens and cost

Question	Input tok	Output tok	Total tok	Cost (USD)
Q1	89	168	257	0.000447
Q2	85	125	210	0.000338
Q3	87	115	202	0.000314
Q4	89	116	205	0.000317
Q5	86	171	257	0.000453

The average cost is USD 0.000374 per request, so 10,000 requests a month would cost about **USD 3.74**. Output tokens are about 93% of the cost, since they are priced 8.3 times higher than input tokens and the answers are longer than the prompts. **Recommendation:** keep the answers short. Add a limit to the prompt (at most 80 words and 3 key points) and set max_tokens to about 250. Reducing the average output from 139 to about 70 tokens would cut the cost per request by about 46% and also reduce latency. Frequently repeated questions can be cached.

4. Task 3 - Prompt versioning with MLflow

The three prompts were registered in the MLflow Prompt Registry as *eduassist_prompt* versions 1-3 with commit messages. v1 is a minimal instruction, v2 adds an academic-assistant role and simple language, and v3 asks for the JSON schema. Each version was loaded from the registry, run on the same five questions and logged as a run in the experiment *eduassist_llmops_assignment*.

Version	Avg Latency	Avg Input Tok	Avg Output Tok	Avg Cost (USD)	Valid JSON	Recommended?
v1	2.179 s	22.2	461.8	0.001161	0%	No
v2	1.947 s	46.2	308.4	0.000785	0%	No
v3	**1.099 s**	**87.2**	**149.4**	**0.000400**	**100%**	**Yes**

Promotion. v3 was given the *production* alias. It is the only version whose output the application can parse: 100% valid JSON, while v1 and v2 returned markdown text (0%). It is also the cheapest and fastest. The prompt is longer than v1, but the schema keeps the answer at 149 output tokens on average instead of 462, so it costs 66% less than v1 and 49% less than v2, and latency is about half of v1. The answers are less detailed than v2, and the confidence value is the model's own estimate. Since the application loads prompts:/eduassist_prompt@production, a new version can be promoted or v3 rolled back by changing the alias.

5. Conclusion

The model answered each question in about 1.2 s for about USD 0.00037. Most of the cost comes from output tokens, so answer length matters for both cost and latency. Comparing the prompt versions in MLflow showed that v3 gives valid JSON and is also the cheapest, so it was promoted. Five questions is a small sample; in production, latency (p95) and the valid JSON rate should be monitored over more requests.